

**A PROPOSED OFFERING OF
A VILLAGE RESIDENT INFORMATION & PROFILING SYSTEM
FOR DISIPLINA VILLAGE, UGONG**

A Project Proposal Presented to the
Faculty of Datamex College of Saint Adeline, Inc.

In Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Information Technology

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TECHNICAL
DOCUMENTATION

INTRODUCTION

The Village Resident Information & Profiling System was created to make managing resident and household records faster, easier, and more accurate for barangay offices. This document explains the technical aspects of the system, including how it was designed, developed, and tested. It serves as a complete guide for users, developers, and maintainers who will use or enhance the system in the future. The main goal is to help readers clearly understand how the system works and how it can be properly configured or fixed when needed.

The system was designed to replace the traditional manual process of recording information through logbooks, index cards, and spreadsheets. These outdated methods often caused delays, data loss, and duplication of records. By digitizing these processes, the system provides a more organized and efficient way to store and manage data. It also reduces human error and ensures that all information is safely kept in a single, secure database.

Through this system, barangay officials can now easily record, update, and retrieve resident information. Admins have full control over approving, editing, and managing data, while staff members assist with verifying and entering resident details. This division of roles helps maintain order and accuracy in the database. It also improves collaboration between users, ensuring that information is always up to date.

This documentation includes all the essential details about the system, from installation and configuration to testing and maintenance. It also explains how each module works and how they connect to form one complete system. The purpose of this document is to guide users in understanding the system's flow and structure. Overall, it ensures that the Village Resident Information & Profiling System remains reliable, efficient, and easy to maintain for future use.

SYSTEM OVERVIEW

The Village Resident Information & Profiling System uses a client-server structure built with VB.NET for the front end and SQL Server Management Studio (SSMS) for the back end. VB.NET handles the forms, buttons, and user interactions, while SSMS manages the storage and retrieval of all resident and household data. Both parts work together to keep data accurate and easy to access.

The system includes several important modules such as Login and Authentication, Resident Management, Household Grouping, Reports, and User Role Control. Each module plays a role in keeping records organized and secure. Admins have full control over data, while staff members handle verification and updates for residents.

The system can be installed and used on any Windows computer within the barangay office. It is designed to run locally, but it can also be connected through a small network if needed. Overall, the system provides a practical and efficient solution for daily record management.

INSTALLATION GUIDE

System Requirements

- Hardware: Intel Core i5 processor, 8GB RAM, 256GB SSD, Windows 10 Pro (64-bit)
- Software: Microsoft Visual Basic.Net 2010, SQL Server Management Studio (SSMS), .NET Framework 4.8

Installation Steps

1. Install Visual Studio 2022 and include VB.NET components during setup.
2. Install SQL Server Management Studio (SSMS) and create a new database named BarangayDB.
3. Import the provided SQL script to create tables and relationships.
4. Open the VB.NET project in Visual Studio and connect it to your local SQL Server database.

5. Build and run the system to verify the setup.

Configuration

If the connection does not work, open the configuration file and update the connection string with your SQL Server name. Make sure the SQL service is running before launching the system. Admin accounts can also be added manually through SSMS if needed.

CONFIGURATION GUIDE

To configure the system, open the App.config file inside your project folder. Update the **connection string** with your SQL Server details, for example:

```
<connectionStrings>  
<add name="BarangayDB" connectionString="Data Source=localhost;Initial  
Catalog=BarangayDB;Integrated Security=True"/></connectionStrings>
```

After saving, rebuild the project in Visual Studio. You can also change the layout, colors, or buttons from the form designer to match your preferred theme. Always make a backup before applying big changes.

For best performance, it's recommended to maintain one admin account and limit access for staff users. This helps prevent data duplication or editing conflicts.

API DOCUMENTATION

The current version of the system does not use any APIs since it runs locally within the barangay office. All operations such as saving, updating, and deleting data are handled directly through SQL queries in VB.NET.

In future versions, APIs can be added if the system is connected to an online or mobile platform. These APIs would include authentication features to keep data secure.

DATABASE DOCUMENTATION

The system's database contains several key tables that store all resident and household data:

- tblResidents – Keeps personal details of all registered residents.
- tblHouseholds – Stores household information and identifies each head of family.
- tblUsers – Manages login credentials and user roles (Admin and Staff).
- tblReports – Keeps records of generated reports for easy access.

Each table is connected using foreign keys to maintain data consistency. Data backups can be made through SSMS Export or by copying the database file. It is recommended to back up data at least once a week or before system updates.

USER MANUAL

When the system starts, users must log in using their assigned username and password. Admins have full access to all system modules, while staff have limited permissions to verify and input resident data. After logging in, the dashboard shows all available options like managing residents, grouping households, and generating reports.

To add a resident, the staff fills out the registration form and submits it for admin approval. The admin can then approve, edit, or delete records as needed. Reports can be generated and printed for barangay documentation.

The interface is simple and easy to navigate. Each button and form is clearly labeled, and helpful messages appear to guide users during data entry.

TROUBLESHOOTING GUIDE

Common Problems and Fixes:

- Cannot connect to database: Make sure SQL Server is running and the connection string is correct.
- Login not working: Check the username and password or reset the admin account in the database.
- Slow system performance: Remove test data or optimize your SQL queries.

If issues continue, restart Visual Studio and SSMS, then rebuild the project. You can also check the error messages in the Visual Studio output window for more details.

CODE DOCUMENTATION

The code is organized into modules such as LoginForm.vb, ResidentForm.vb, HouseholdForm.vb, and ReportForm.vb. Each form contains clear and descriptive naming for buttons, labels, and variables.

Inline comments are used throughout the code to explain the purpose of important functions. This makes it easier for future developers to understand and update the system.

All code follows standard VB.NET formatting rules for readability and consistency.

TESTING DOCUMENTATION

The system went through multiple rounds of testing to ensure that all features worked properly. Manual and user acceptance testing were done by both the admin and staff to simulate real scenarios. Each module, including login, registration, and resident management, passed all major test cases successfully.

All bugs found during testing were logged and fixed before final deployment. The final results confirmed that the system is stable and ready for use.

MAINTENANCE GUIDE

Regular maintenance keeps the system stable and reliable. Always back up the database weekly using SSMS to prevent data loss. Any system update or modification should be tested first before applying it to the live version.

For code updates, use version control tools like GitHub to track changes and avoid overwriting files. Document every change or improvement made to help future developers understand what was modified.

REVISION HISTORY

Version	Date	Description	Updated By
1.0	October 2025	Initial release of system documentation	Carlo Factor
1.1	October 2025	Added testing and user feedback	Carlo Factor

APPROVAL

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APPENDIX

The appendix includes extra references that support this documentation.

Screenshots of the system interface

Database design (ERD)

Test results and bug reports

User feedback forms